Cube Satellite

For Earth Observation



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01

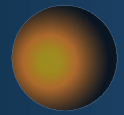
Survey

Satellite types, uses and orbits

What is Satellite

A satellite is an object that orbits a planet, such as Earth, or a star. There are many different types of satellites, including artificial satellites, which are made by humans and sent into space for a specific purpose

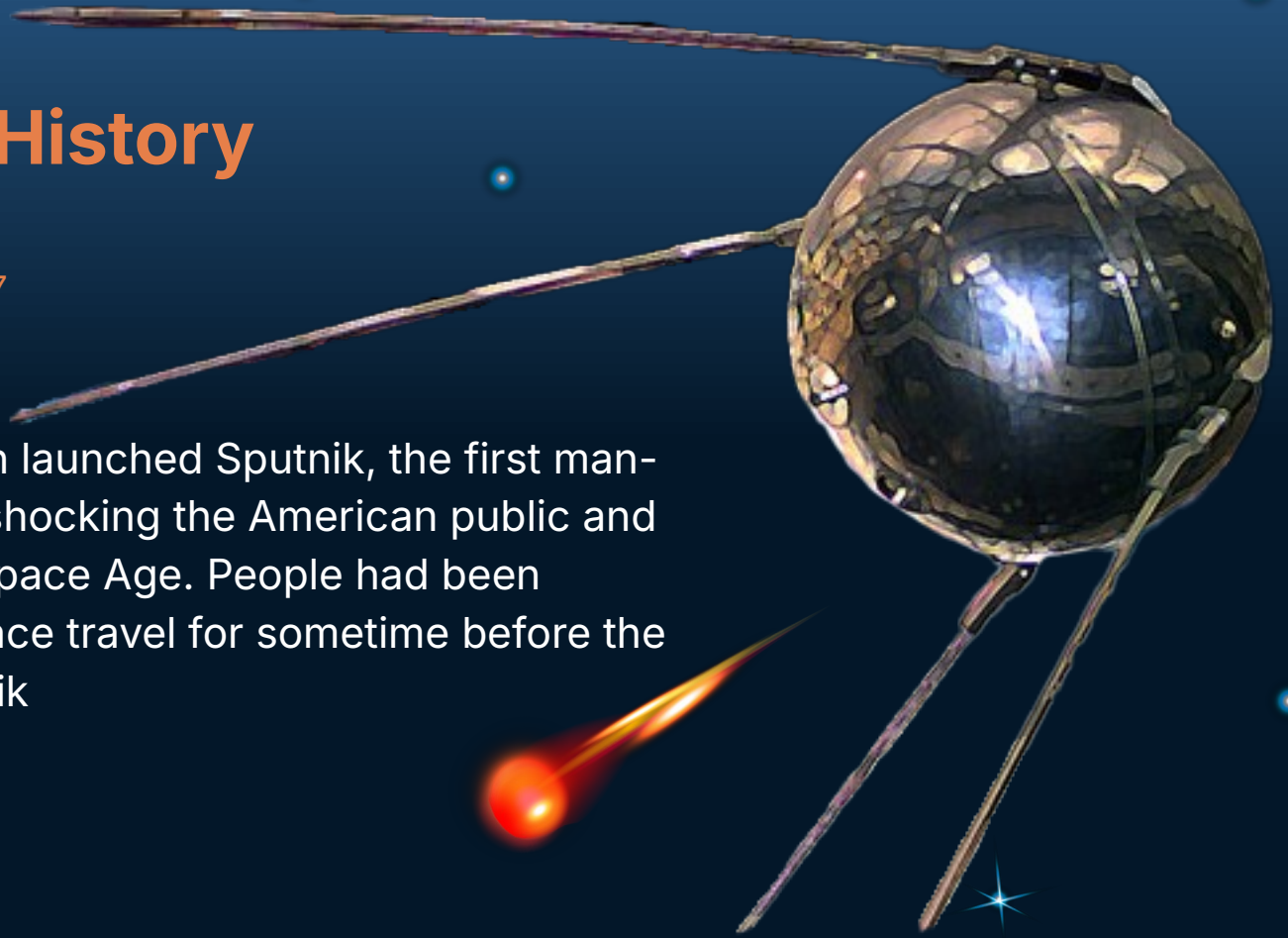




Satellite History

On October 4, 1957

the Soviet Union launched Sputnik, the first man-made satellite, shocking the American public and beginning the Space Age. People had been dreaming of space travel for sometime before the launch of Sputnik



Power Unit :
Most satellites rely on a solar array to convert sunlight into energy.

Thermal Control:
Guards satellite equipment against extreme changes in temperature.

Housing:
Constructed from strong materials that can stand the harsh space environment.

Transponders:
Uplink and downlink signals arrive and depart at different frequencies. Transponders convert uplinked frequencies to downlink frequencies and then amplify the converted Signal for sending to Earth.

Antenna
Used to receive and transmit Signal from Earth.

Guidance and Stabilization:
Sensors monitor the satellite's position to ensure it remains in the correct orbit and is oriented toward the correct target. If necessary, thrusters and other maneuvers allow a satellite to fine-tune its position and orientation.

Command and Data Handling:
The operational heart of a satellite, command and control systems, Receive commands from Ground Station.

Low Earth Orbit (LEO)

refers to the region of Earth's orbit closest to the planet's surface, typically defined as an altitude of up to 2,000 km

Medium Earth Orbit (MEO)

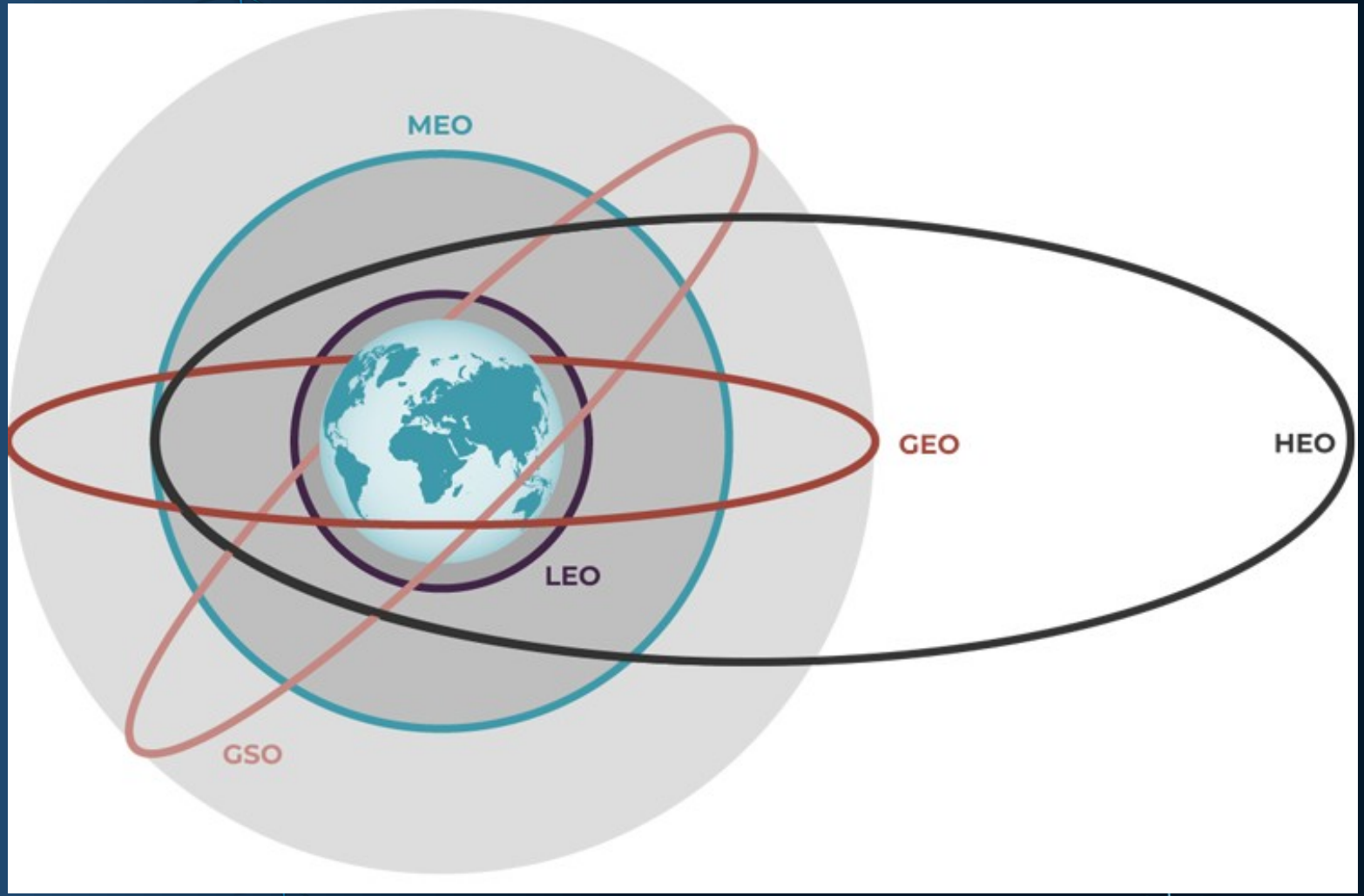
is the region of space located between low Earth orbit (LEO) and geostationary orbit (GEO). MEO is typically defined as an altitude range of 20,200 km to 35,786 km above the Earth's surface

Geostationary Earth Orbit

is a type of orbit located approximately 36,000 kilometers (22,000 miles) above the Earth's equator

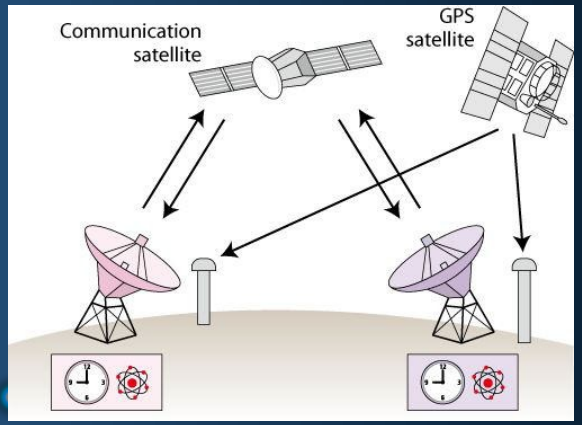
High Earth Orbit

High Earth orbit refers to orbits above the atmosphere, typically at altitudes of thousands of kilometers


MEO
GEO
HEO
LEO
GSO

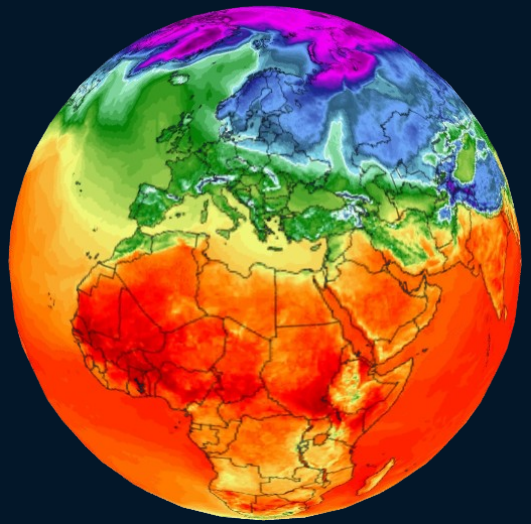
Communication

These satellites are used for telecommunications, such as phone and internet services

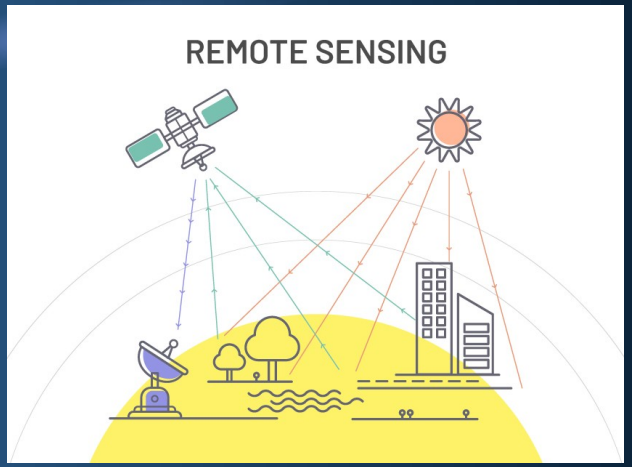


Weather

These satellites are used to monitor weather patterns and provide forecasts

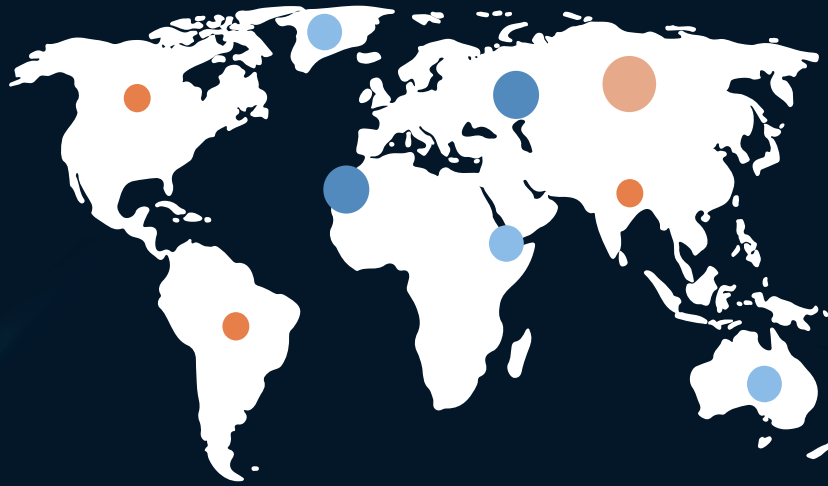


Remote sensing satellites



These satellites are used to gather information about the Earth's surface, such as land use, vegetation, and water resources.

Navigation satellites



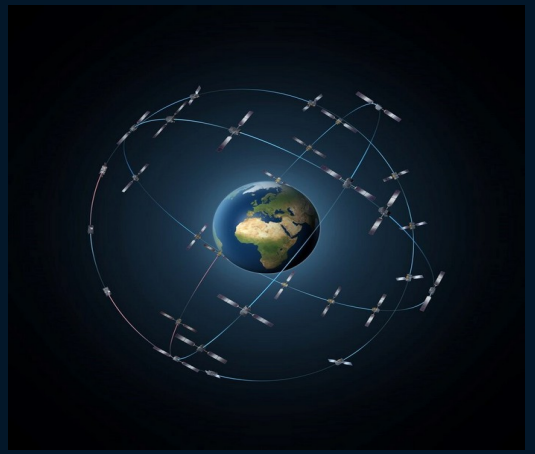
These satellites are used for GPS and other navigation services.

Military



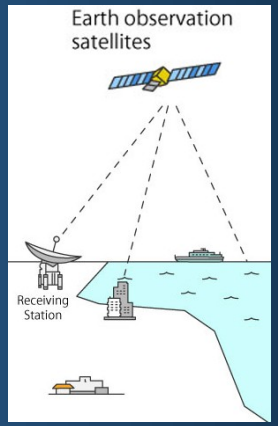
These satellites are used for military intelligence and surveillance

Scientific research



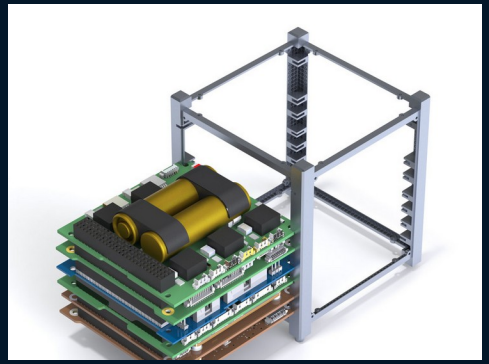
These satellites are used for scientific research and exploration

Earth Observation



These satellites are used to observe and measure different aspect of earth

CubeSat



Small satellites built from 10 cm cube units



Broadcasting

Satellites are used to broadcast television and radio programs to a wide area



Telecommunication

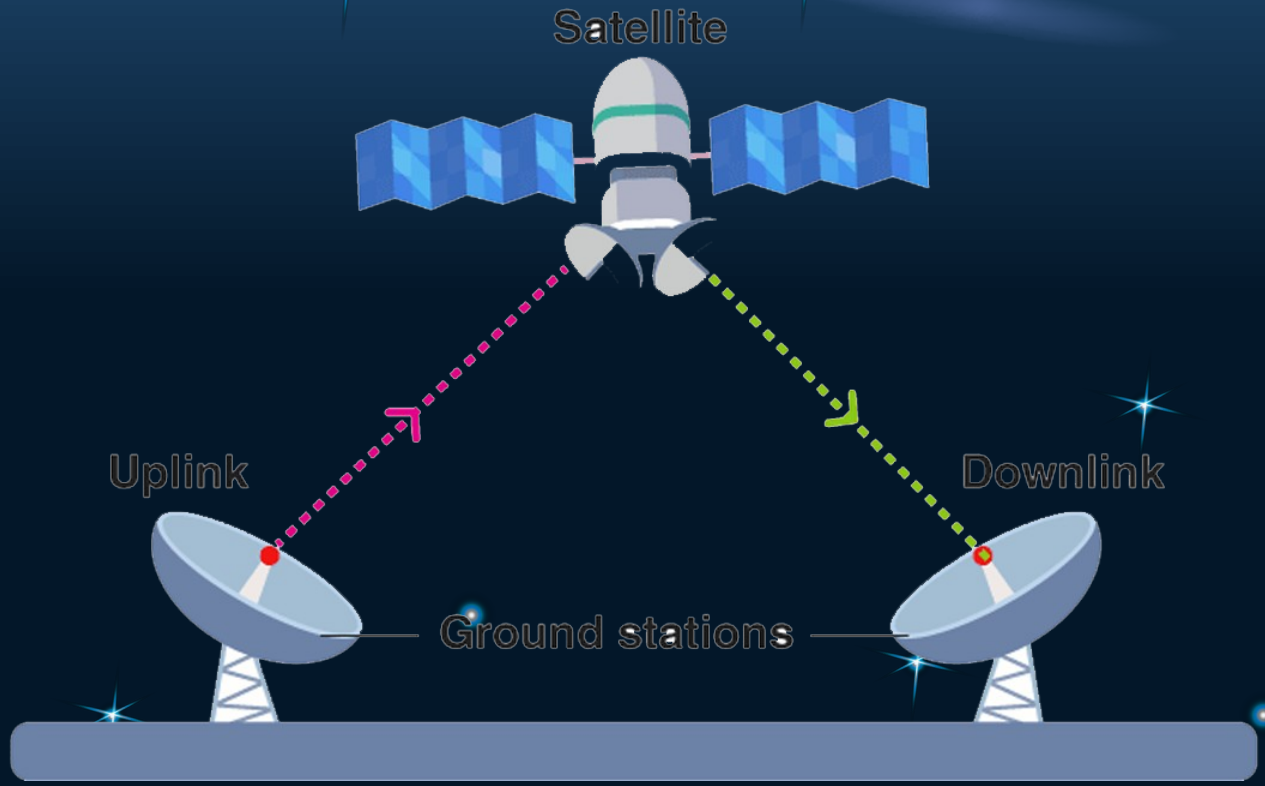
Communication satellites are used to provide phone, internet, and television services to remote and hard-to-reach areas.



Location services

Satellites are used for GPS and other navigation services, such as tracking vehicles, ships, and aircraft.

Satellite work principle



Satellite Stability

there are two main forces that act on a satellite in space:
the force of gravity and the centrifugal force.
The balance between these two forces is what allows the satellite to maintain a stable orbit. If the force of gravity were to become stronger, the satellite would be pulled closer to the Earth, and if the centrifugal force were to become stronger, the satellite would be pushed further away from the Earth. By maintaining the right balance between these two forces, the satellite can remain in a stable orbit and perform its intended function. 2. The force of gravity is the force that attracts the satellite towards the center of the Earth. This force is responsible for keeping the satellite in orbit around the Earth. 1. The centrifugal force is the force that pushes the satellite away from the center of the Earth. This force is caused by the satellite's motion in orbit, as it is constantly moving in a circular path around the Earth

Examples

Scientific Satellite for astronomy and cosmology

JWST

High Image Resolution of the earth’s surface

Landsat 8



Milstar

Military communication satellite serves armed force

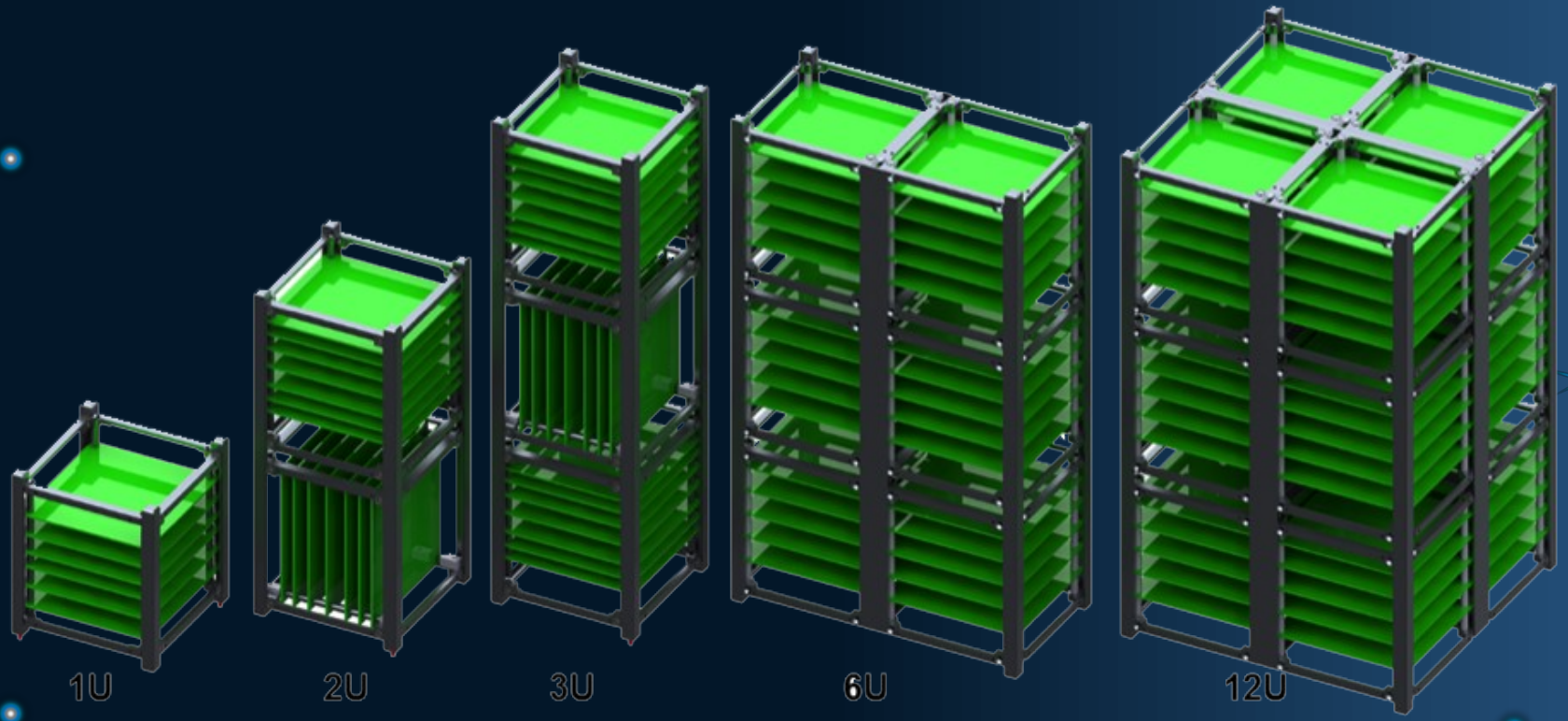
The Iridium

Global voice and data communication

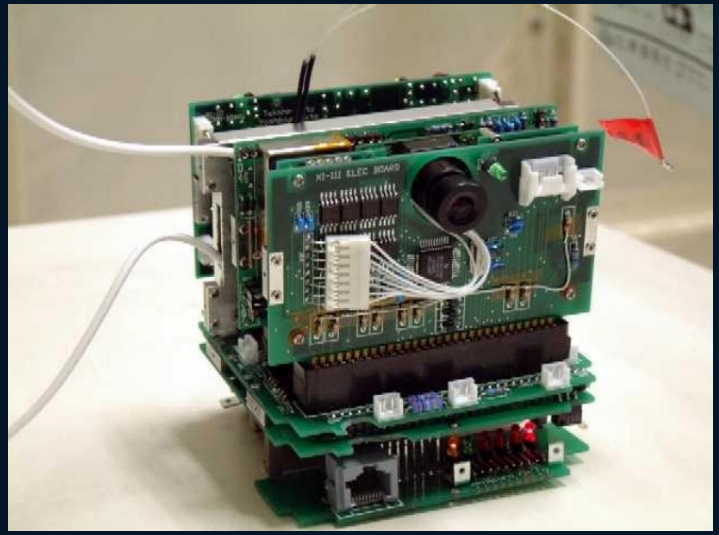
CubeSat

A CubeSat is a type of miniaturized satellite made up of multiples of 10cm cube units, typically weighing between 1 and 10 kg. These small satellites are low-cost to design, build, and launch, making them accessible to a wide range of organizations. They are often used for earth observation, remote sensing, communications, and technology demonstrations and are frequently launched as secondary payloads on larger rockets. CubeSat's have gained popularity as a cost-effective way for organizations to gain experience in satellite development and operations.

CubeSat Types



CubeSat's have become popular among universities due to their cost-effectiveness, standardization, and easy access to launch opportunities. They provide students with hands-on experience in various fields such as aerospace engineering, computer science, and more. They also offer a platform for universities to conduct research in earth observation, atmospheric studies, and space environment monitoring. CubeSat's are relatively inexpensive, standardized and accessible for university research and student training.



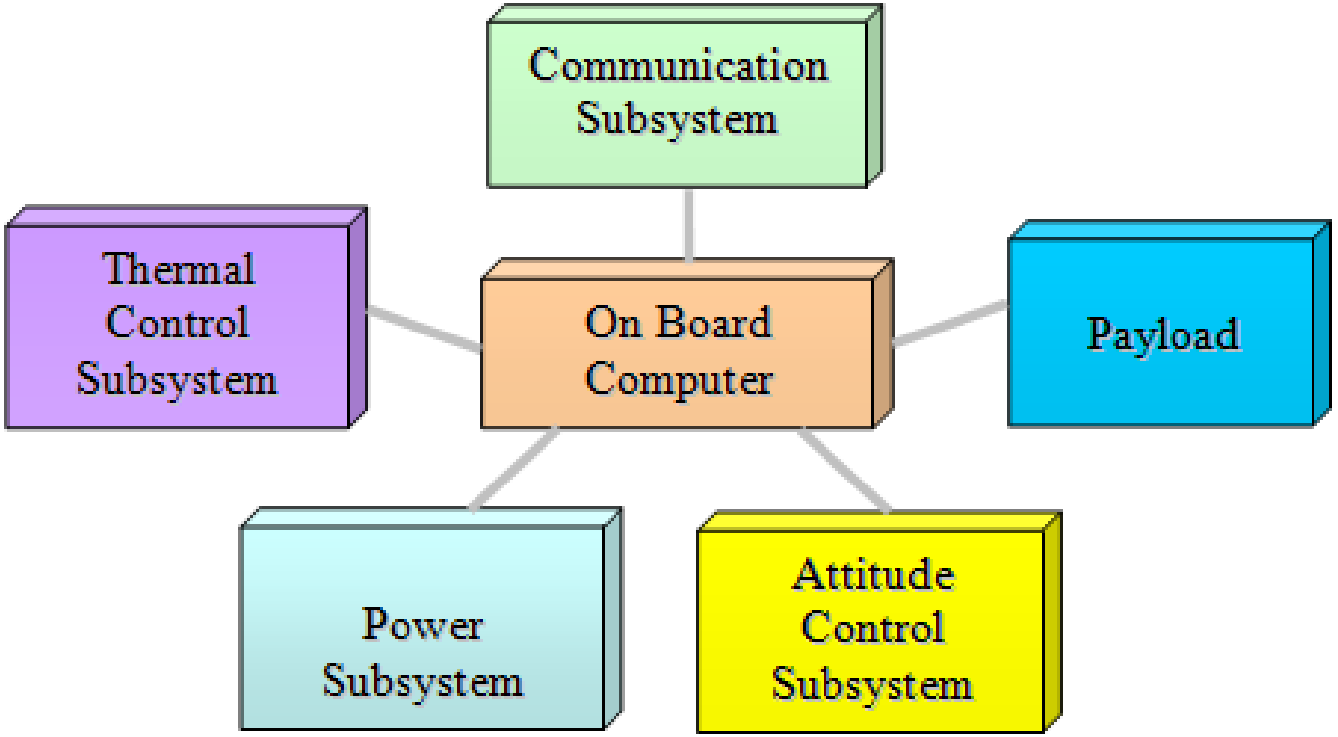
CubeSat payloads refer to the specific instruments that a CubeSat is designed to operate. These payloads can include a wide range of scientific and technological instruments, such as:

- Cameras for imaging
- Magnetometers for space environment monitoring
- Radio transceivers for communication
- GPS receivers for navigation
- Sensors for environmental monitoring
- Scientific experiments
- Commercial payloads for earth observation and remote sensing

So CubeSat payloads can be designed for a variety of different missions

CubeSat Mission

Capture High-resolution images of earth
Temperature and humidity data
Self-Charge itself using Solar Power
Measure different space gases values.


Communication Subsystem
Thermal Control Subsystem
On Board Computer
Payload
Power Subsystem
Attitude Control Subsystem

1

Attitude determination and control system: which includes sensors, actuators, and control algorithms

2

On-board system: which includes computer, memory, and data storage

3

Payload system: which includes the specific instruments that the CubeSat is designed to operate

4

The thermal control subsystem:
it maintains optimal temperature inside the CubeSat using sensors, insulation, heaters and radiators, controlled by software algorithms.

5

Power system:
which includes solar panels, batteries, and power management units

6

Communications system:
which includes antennas, transceivers, and modems

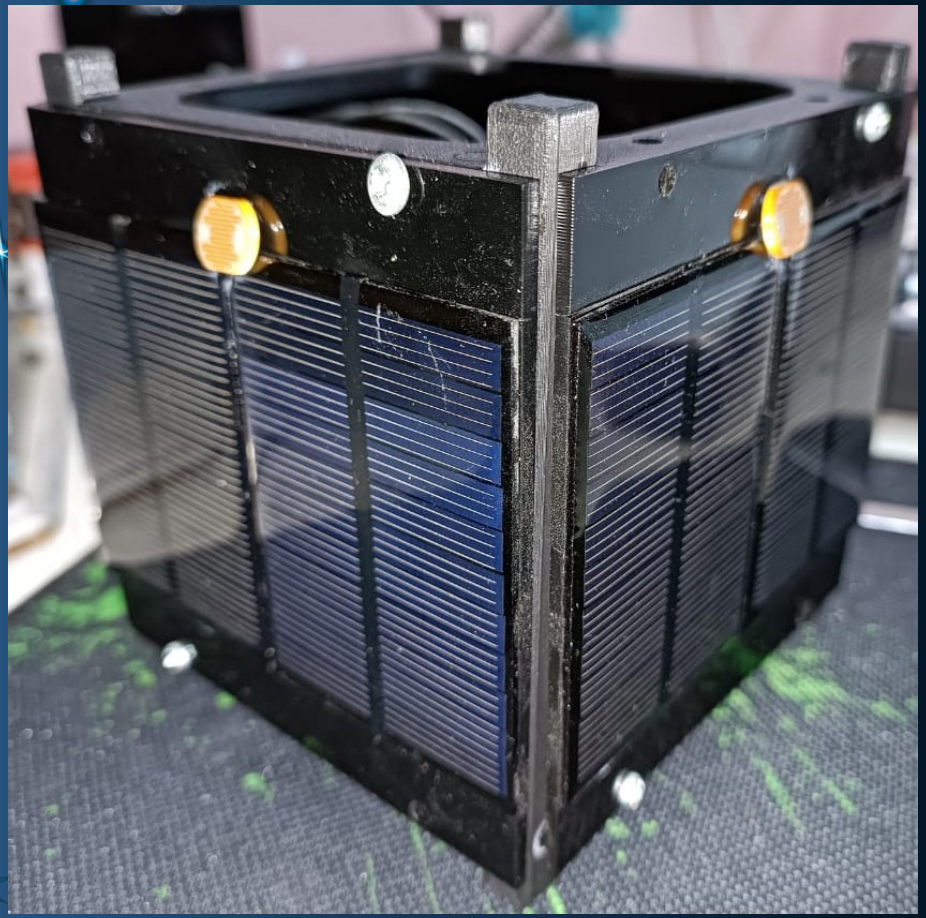


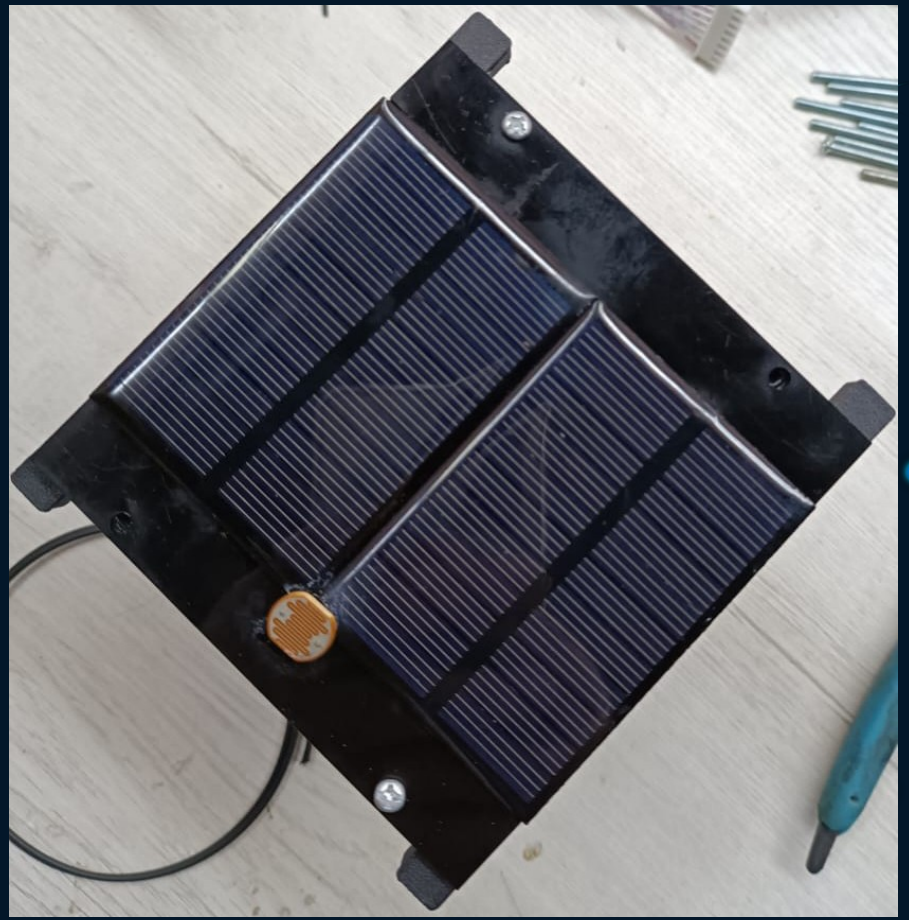
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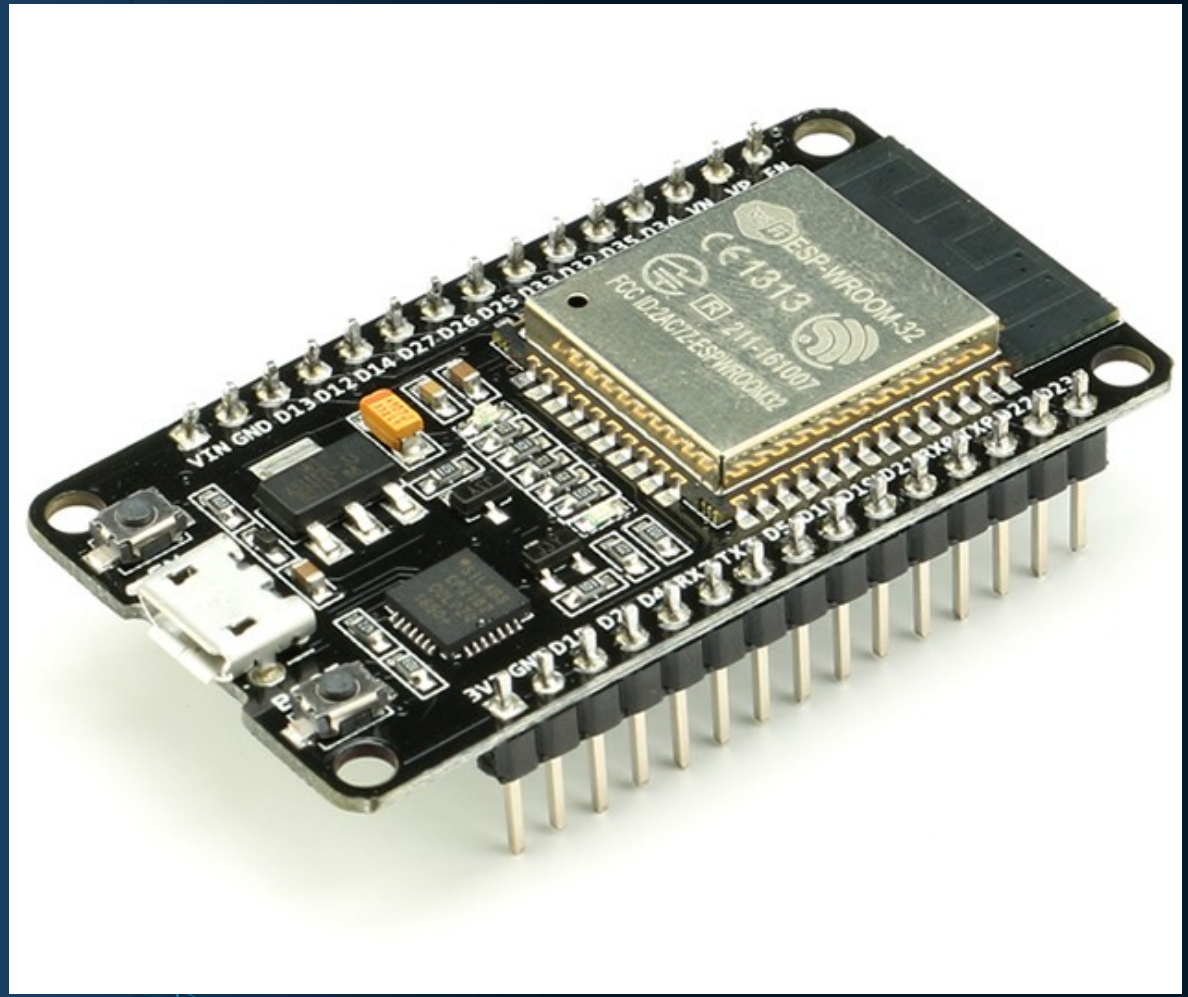
Component

The parts we used to build our CubeSat

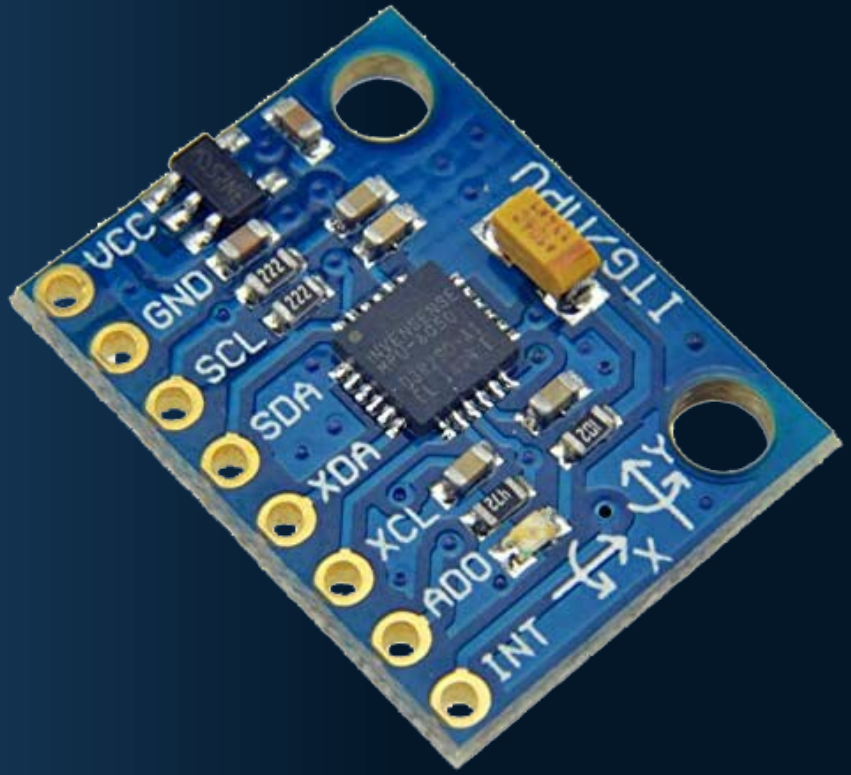


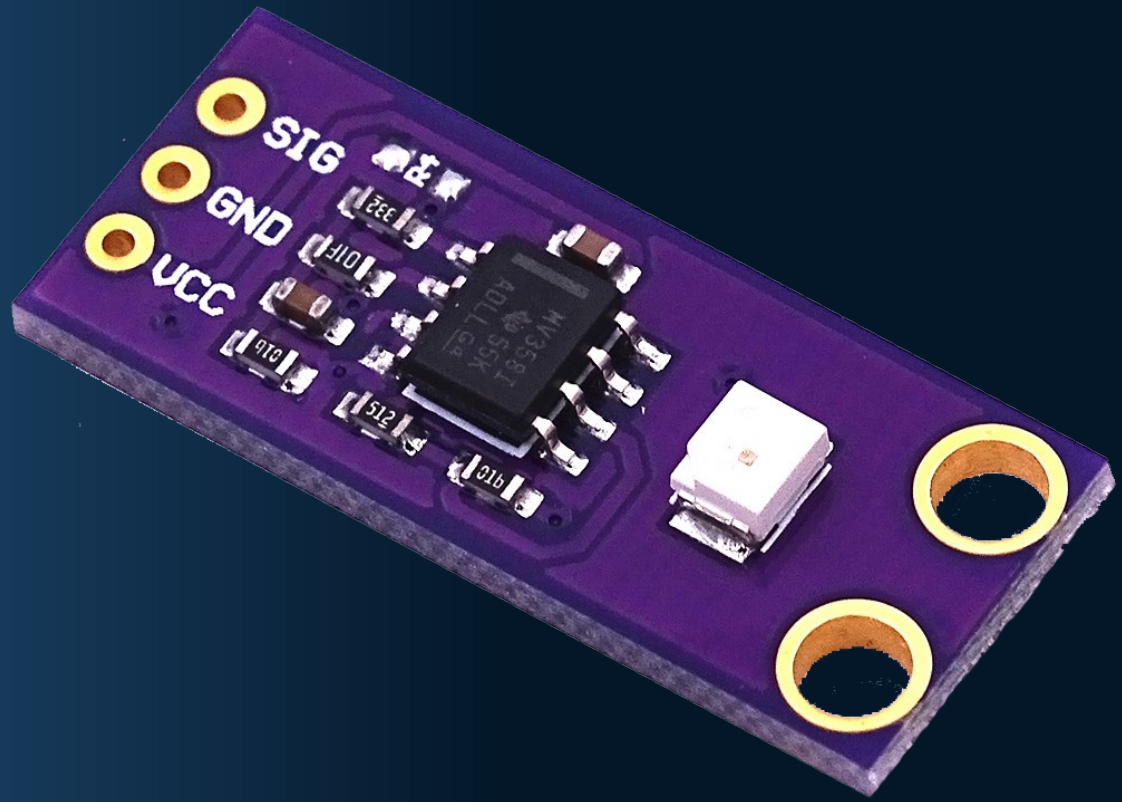





VIN GND D13 D12 D14 D27 D26 D25
ESP-WROOM-32
1313
3V GND D15 D2 D4 RX2 TX2

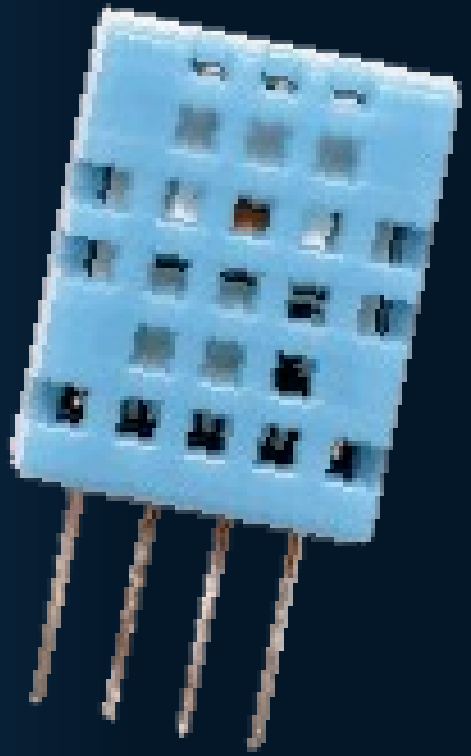

GY-GPS6MV2
VCC RX TX GND
NEO-6M-0-000
1308
0200 24


VCC
GND
SCL
SDA
XDA
XCL
ADO
INT
Y
X

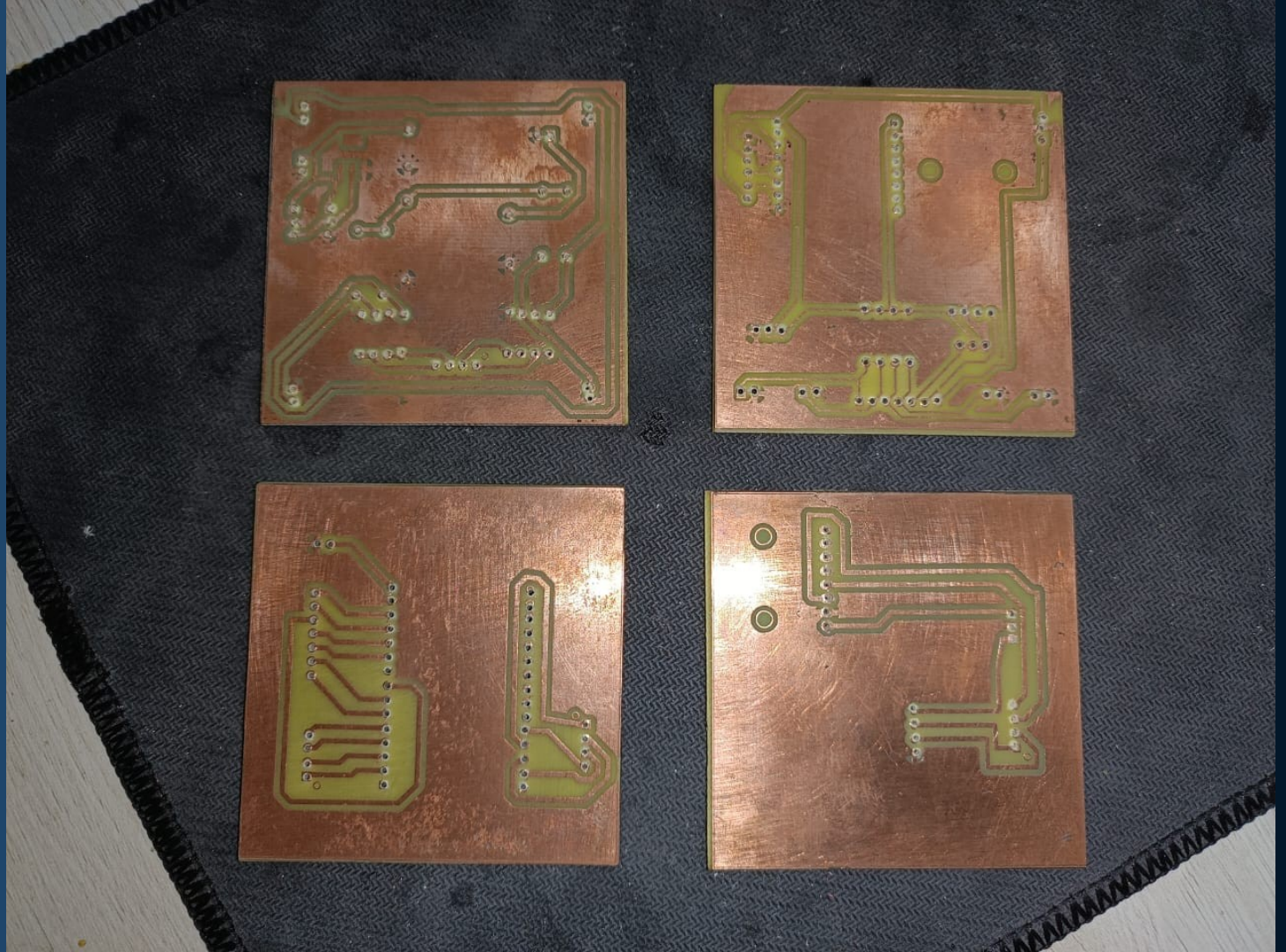

SIG
GND
VCC


ESP32-CAM
Waveshare


Flying-Fish

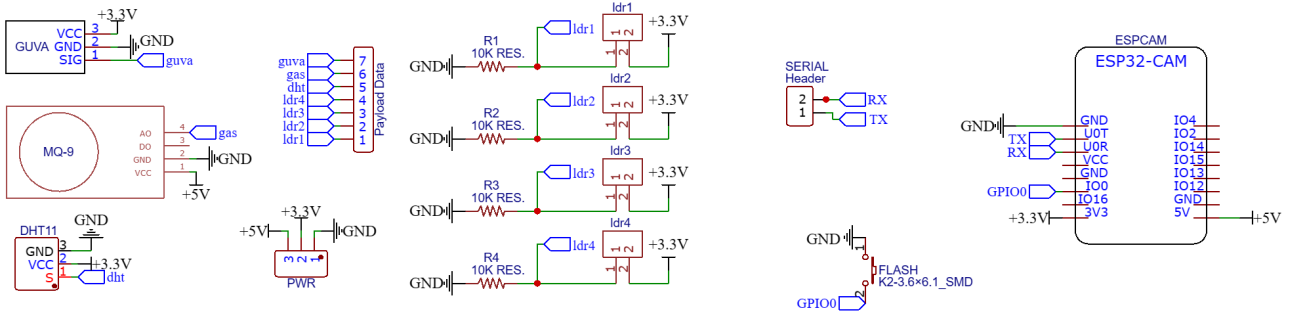



18650-3.7V-2200mAh
2019/05/02

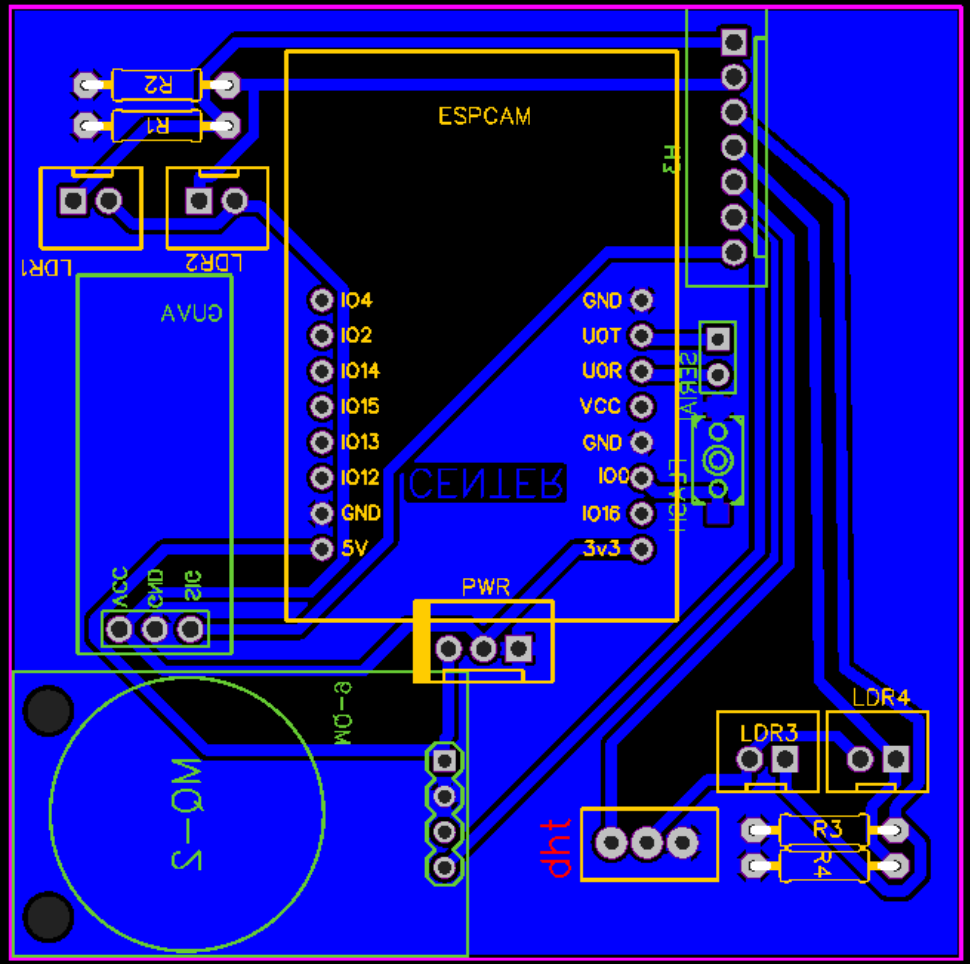


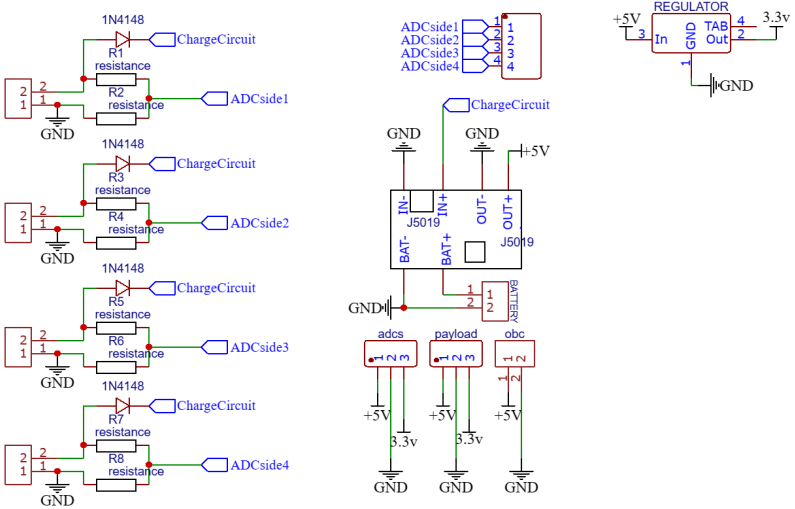
CubeSat

Circuit Scheme

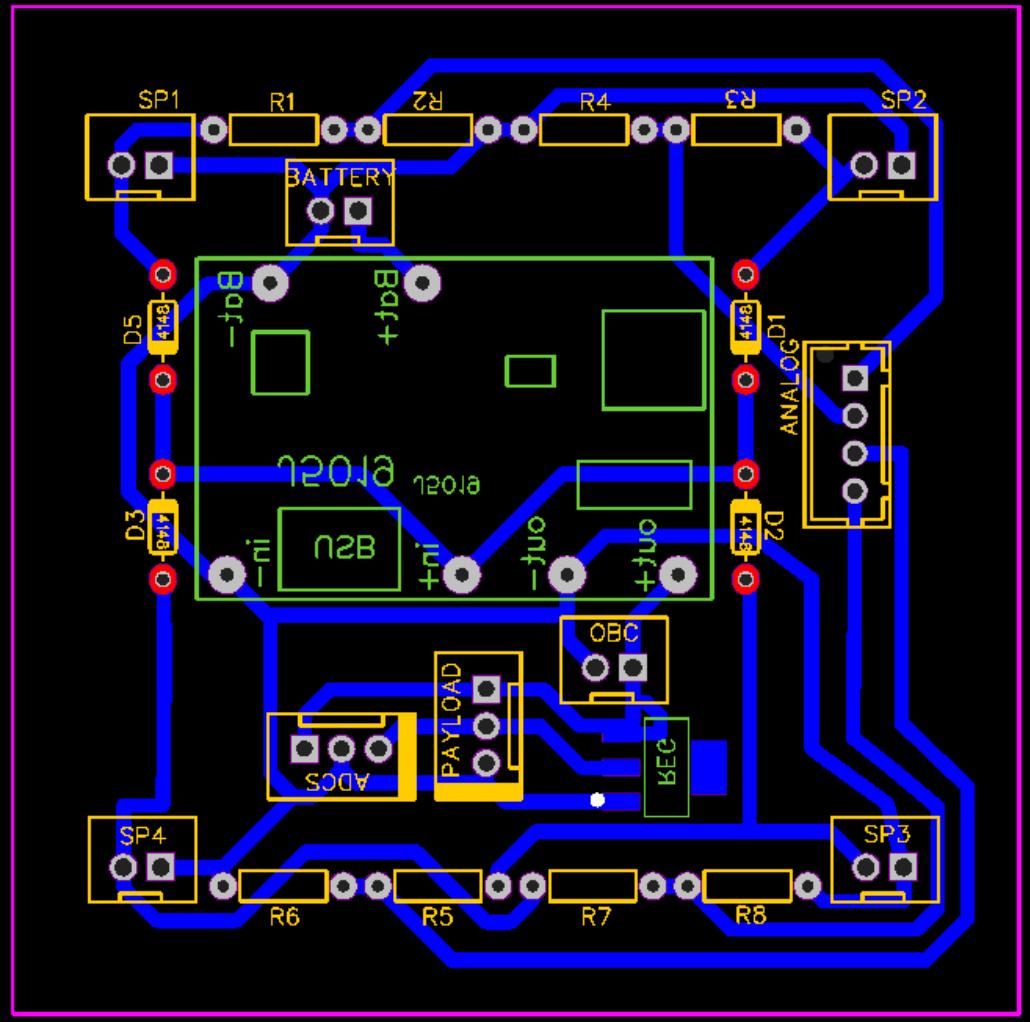


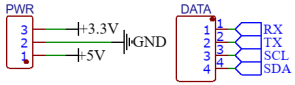
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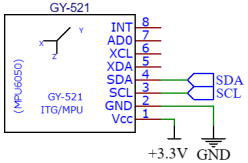

ESPCAM
R2
R1
LDR1
LDR2
AVUG
IO4
IO2
IO14
IO15
IO13
IO12
GND
5V
GND
UOT
UOR
VCC
GND
IOO
IO16
3v3
CENTER
PWR
ACC
GND
SIG
MQ-9
S-QM
LDR3
LDR4
dht
R3

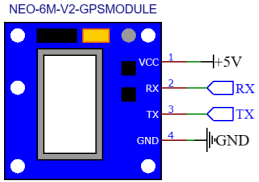


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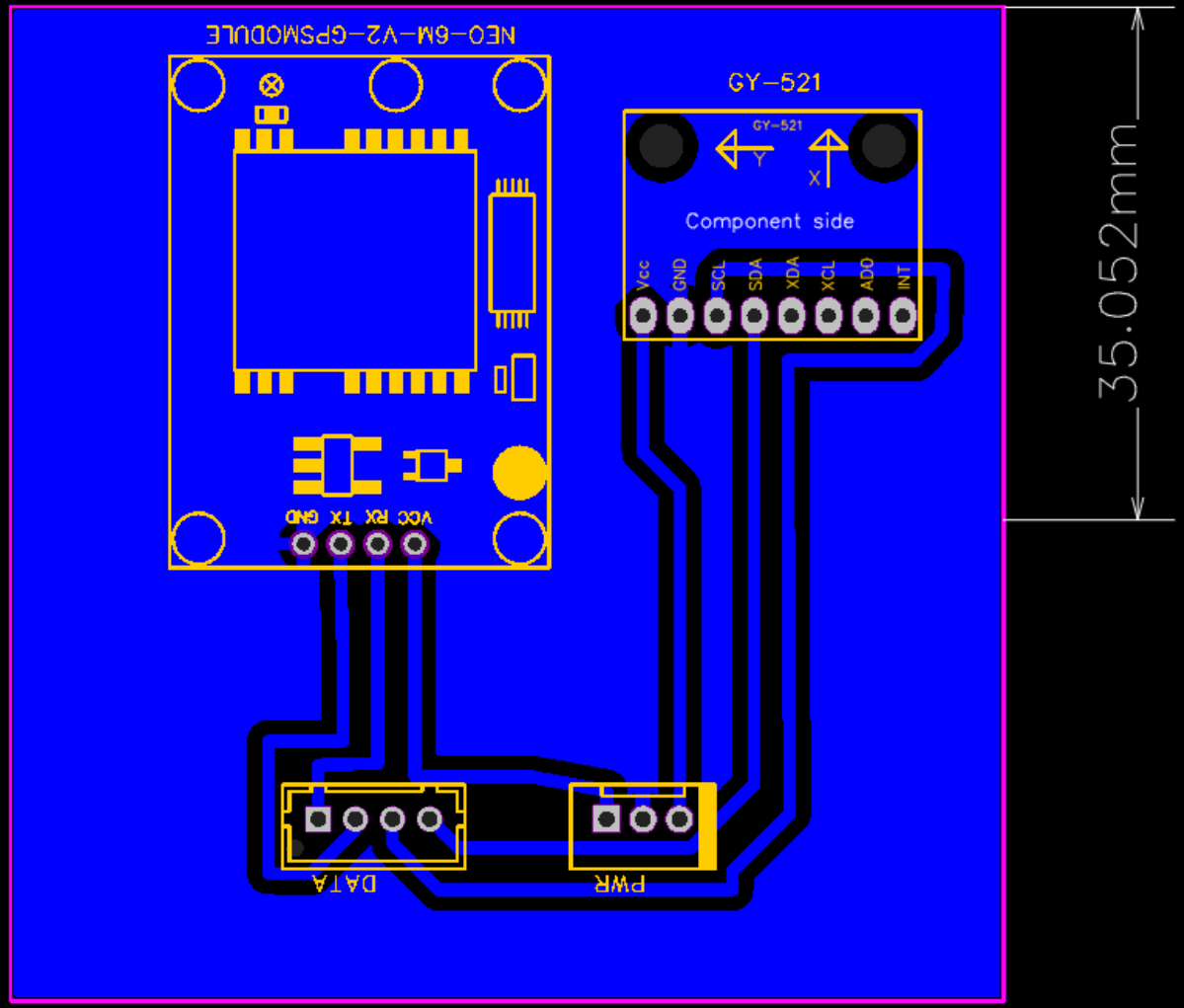

SP1
R1
R2
R4
R3
SP2
BATTERY
D5
D1
ANALOG
D3
D2
OBC
PAYLOAD
SP4
SP3
R6
R5
R7
R8

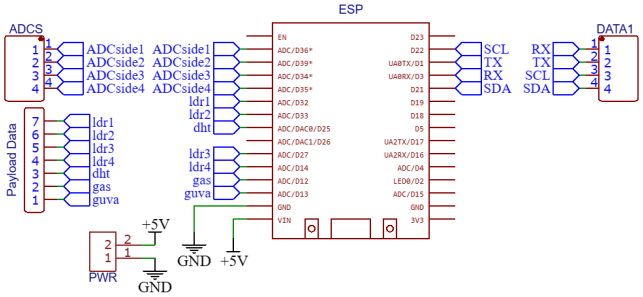






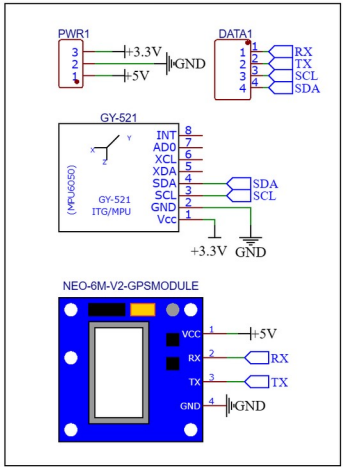
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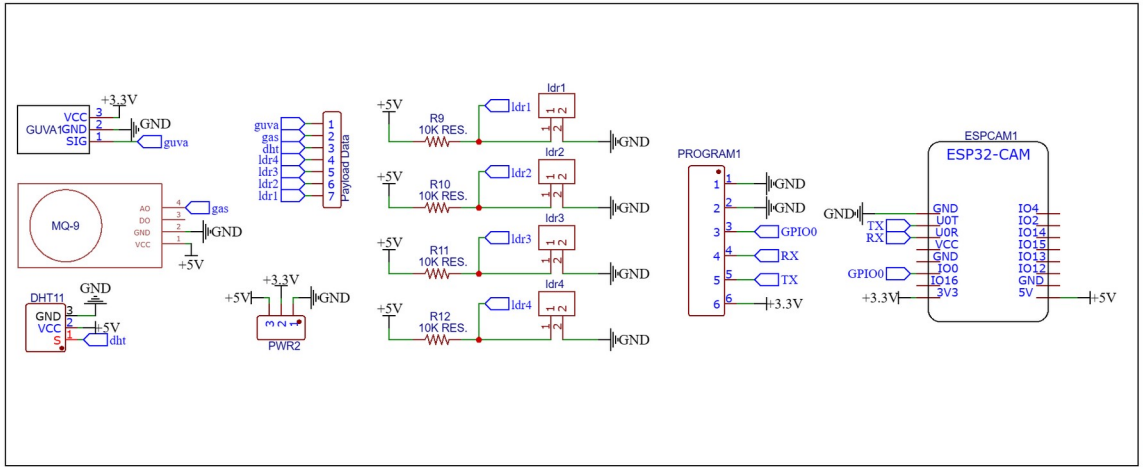

NEO-6M-V2-GPSMODULE
GY-521
GY-521
Y
X
Component side
Vcc
GND
SCL
SDA
XDA
XCL
ADO
INT
GND TX RX VCC
DATA
PWR
35.052mm

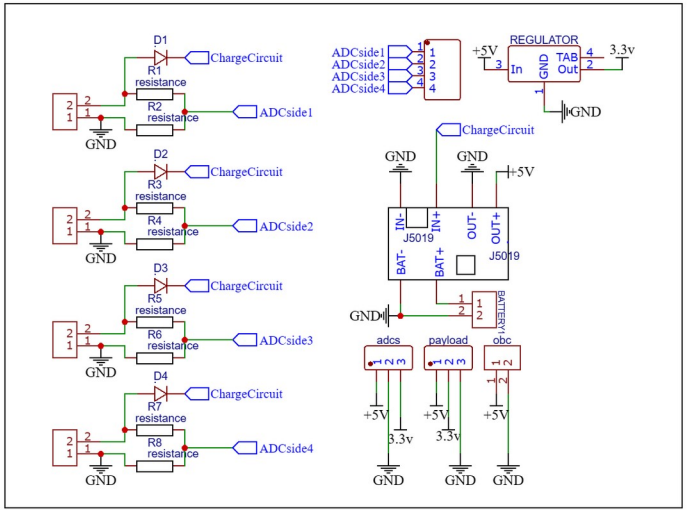


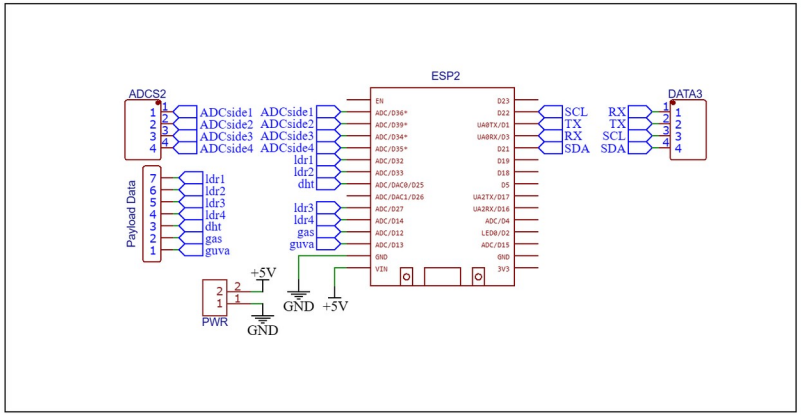
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ESP
D23
D22
TX0
RX0
D21
D19
D18
D05
TX2
RX2
D04
D02
D15
GND
3V3
EN
D36
D39
D34
D35
D32
D33
D25
D26
D27
D14
D12
D13
GND
VIN
ADCS
ADC
PAYLOAD
PWR
Boot
USB
Enable



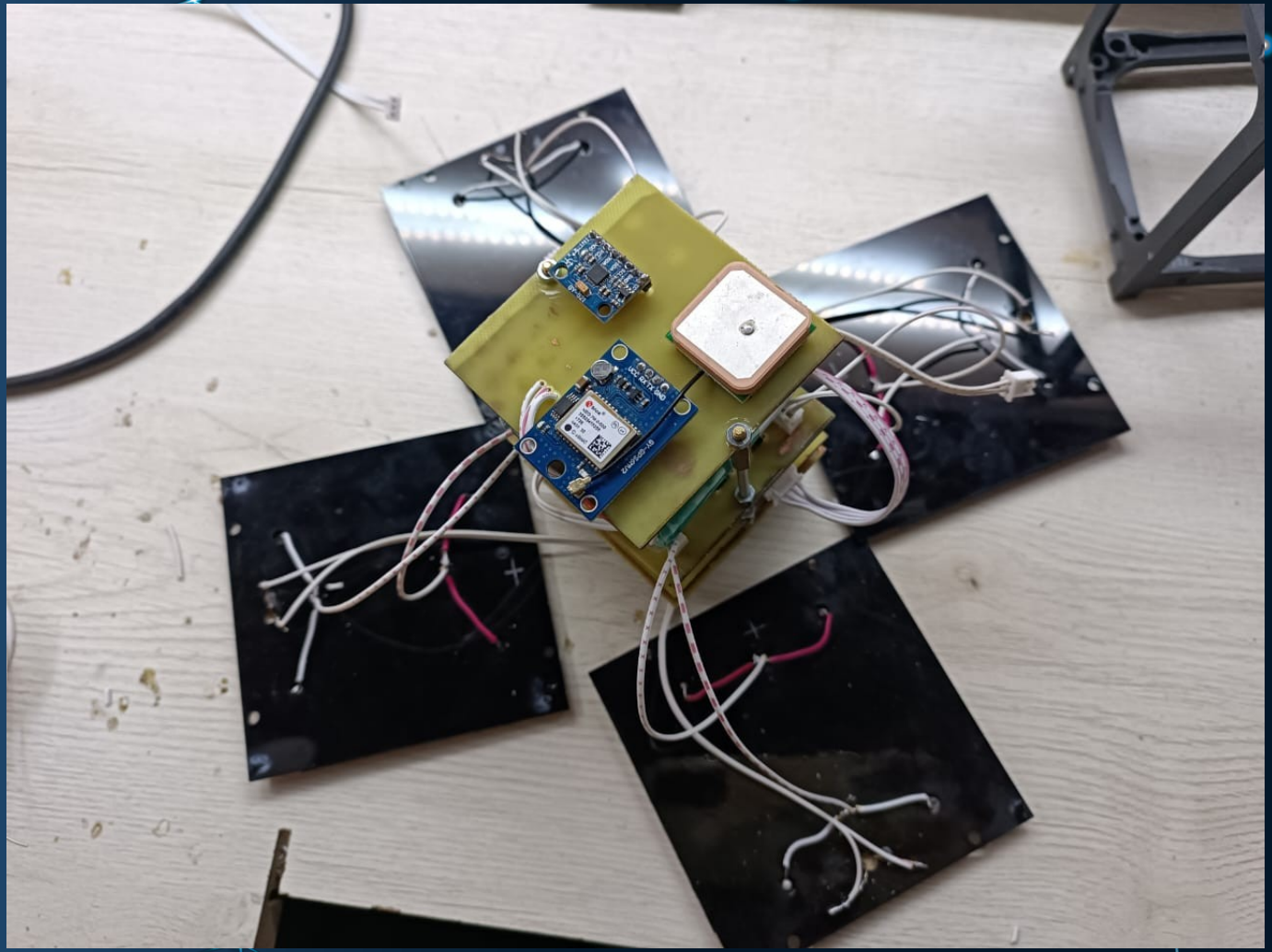


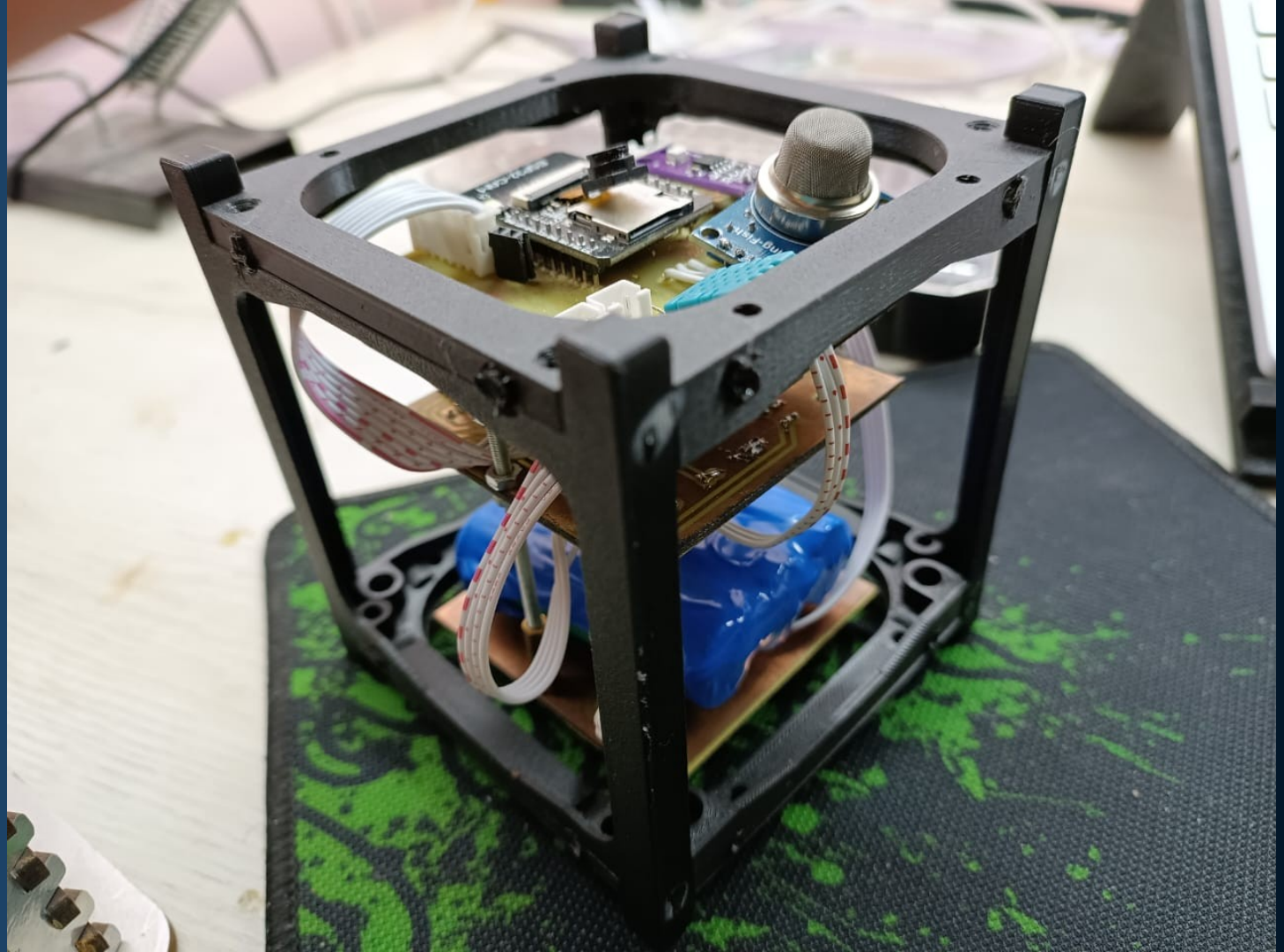


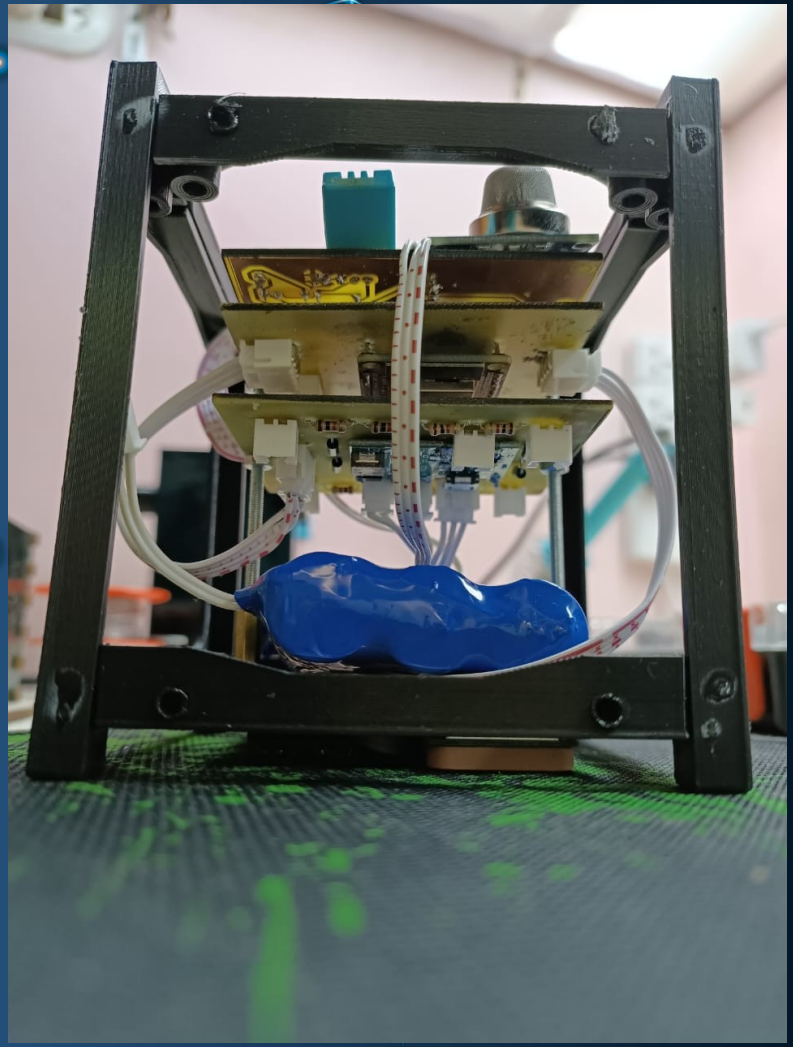


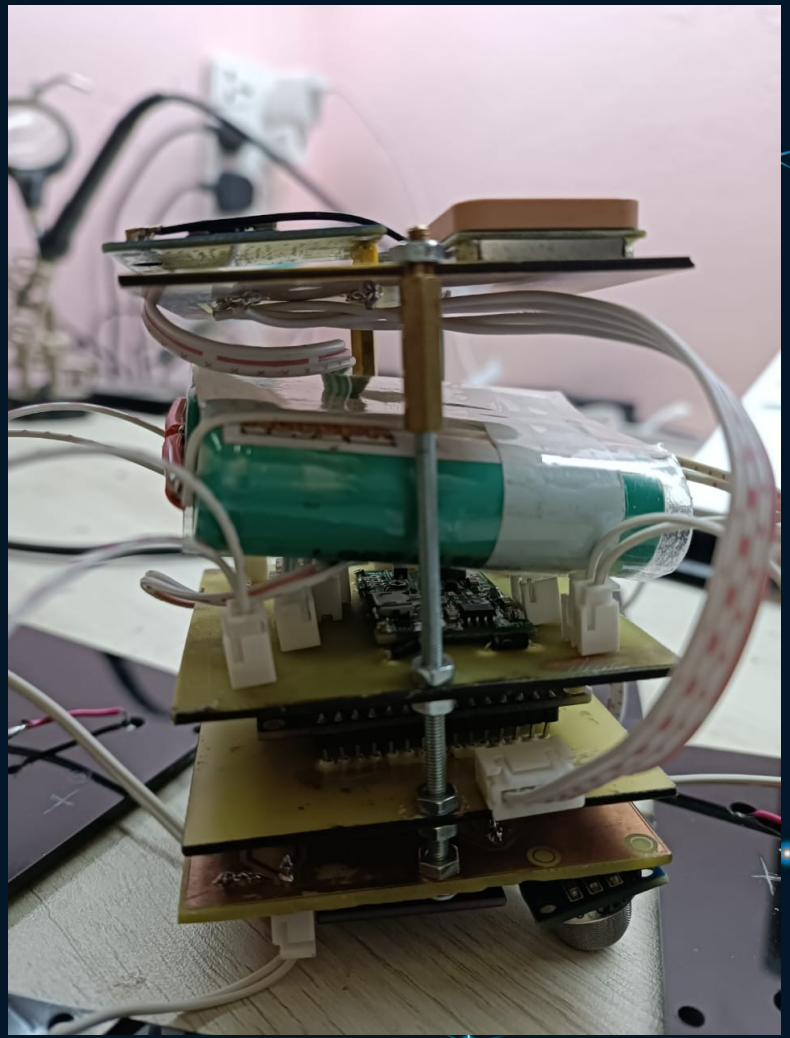
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Circuit PCB



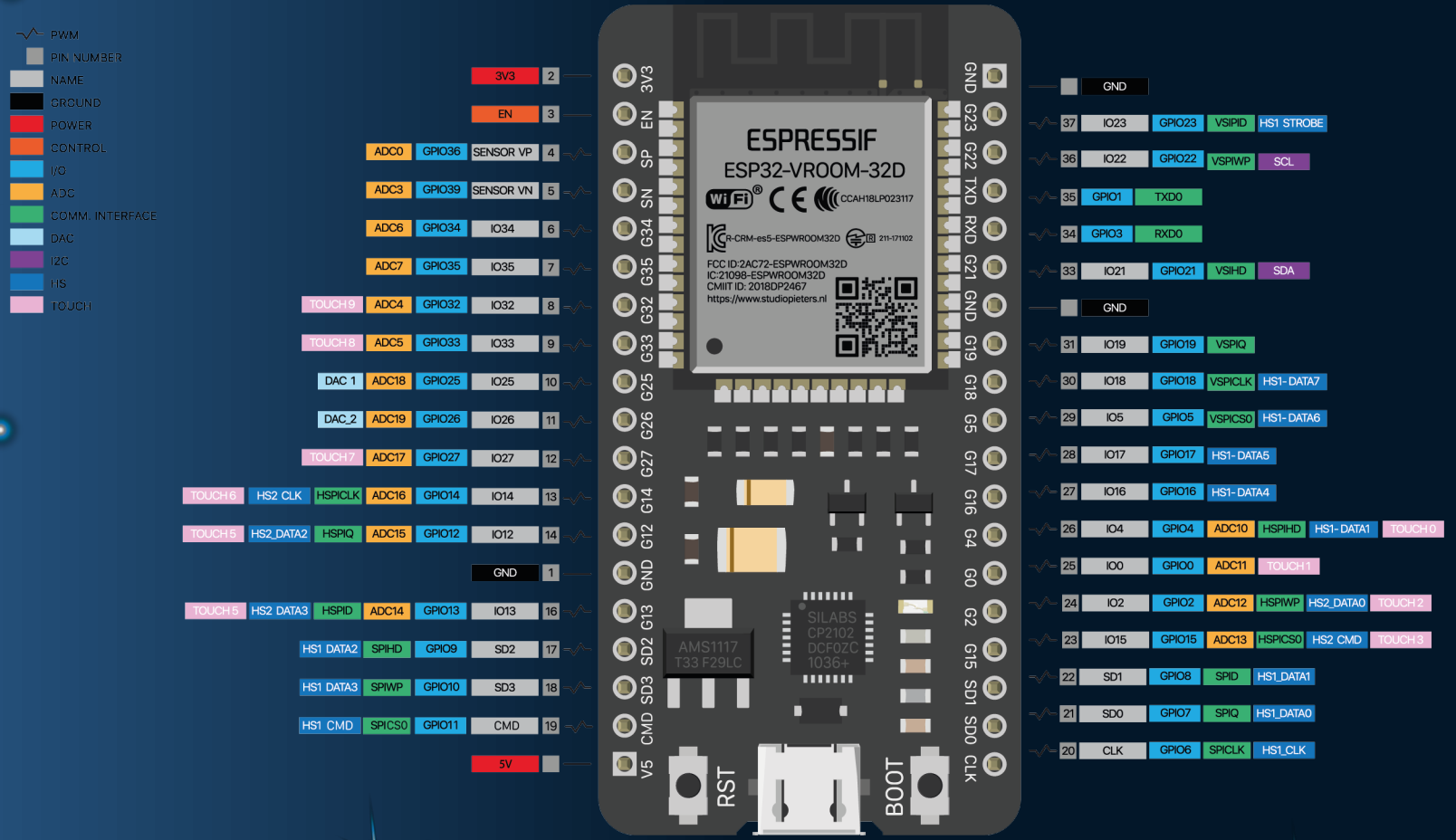







ESP32-CAM
Flying-Fish

Coding


PWM
PIN NUMBER
NAME
GROUND
POWER
CONTROL
I/O
ADC
COMM. INTERFACE
DAC
I2C
HS
TOUCH
3V3 2
EN 3
ADC0 GPIO36 SENSOR VP 4
ADC3 GPIO39 SENSOR VN 5
ADC6 GPIO34 IO34 6
ADC7 GPIO35 IO35 7
TOUCH9 ADC4 GPIO32 IO32 8
TOUCH8 ADC5 GPIO33 IO33 9
DAC 1 ADC18 GPIO25 IO25 10
DAC_2 ADC19 GPIO26 IO26 11
TOUCH7 ADC17 GPIO27 IO27 12
TOUCH6 HS2 CLK HSPICLK ADC16 GPIO14 IO14 13
TOUCH5 HS2_DATA2 HSPIQ ADC15 GPIO12 IO12 14
GND 1
TOUCH5 HS2 DATA3 HSPID ADC14 GPIO13 IO13 16
HS1 DATA2 SPIHD GPIO9 SD2 17
HS1 DATA3 SPIWP GPIO10 SD3 18
HS1 CMD SPICS0 GPIO11 CMD 19
5V
3V3 EN SP SN G34 G35 G32 G33 G25 G26 G27 G14 G12 GND G13 SD2 SD3 CMD V5
GND G23 G22 TXD RXD G21 GND G19 G18 G5 G17 G16 G4 G0 G2 G15 SD1 SD0 CLK
ESPRESSIF
ESP32-VROOM-32D
CCAH18LP023117
R-CRM-es5-ESPWROOM32D
211-171102
FCC ID:2AC72-ESPWROOM32D
IC:21098-ESPWROOM32D
CMIIT ID: 2018DP2467
https://www.studiopieters.nl
AMS1117
T33 F29LC
SILABS
CP2102
DCF0ZC
1036+
RST
BOOT
GND
37 IO23 GPIO23 VSIPID HS1 STROBE
36 IO22 GPIO22 VSPIWP SCL
35 GPIO1 TXD0
34 GPIO3 RXD0
33 IO21 GPIO21 VSIHD SDA
GND
31 IO19 GPIO19 VSPIQ
30 IO18 GPIO18 VSPICLK HS1- DATA7
29 IO5 GPIO5 VSPICS0 HS1- DATA6
28 IO17 GPIO17 HS1- DATA5
27 IO16 GPIO16 HS1- DATA4
26 IO4 GPIO4 ADC10 HSPIHD HS1- DATA1 TOUCH 0
25 IO0 GPIO0 ADC11 TOUCH 1
24 IO2 GPIO2 ADC12 HSPIWP HS2_DATA0 TOUCH 2
23 IO15 GPIO15 ADC13 HSPICS0 HS2 CMD TOUCH 3
22 SD1 GPIO8 SPID HS1_DATA1
21 SD0 GPIO7 SPIQ HS1_DATA0
20 CLK GPIO6 SPICLK HS1_CLK

Our ESP32 code

Reads all sensors every 2 seconds:

- DHT11: temperature and humidity
- MPU-6050: gyroscope and accelerometer
- MQ-9 gas sensor and GUVA-S12SD UV sensor
- 4 LDR sun sensors (one per side)
- NEO-6M GPS: position, altitude, UTC time

Sends live readings over Wi-Fi to a web page that refreshes every 2 seconds

Code: github.com/moustafagabbas/cubesat-1u-earth-observation

Thanks!

